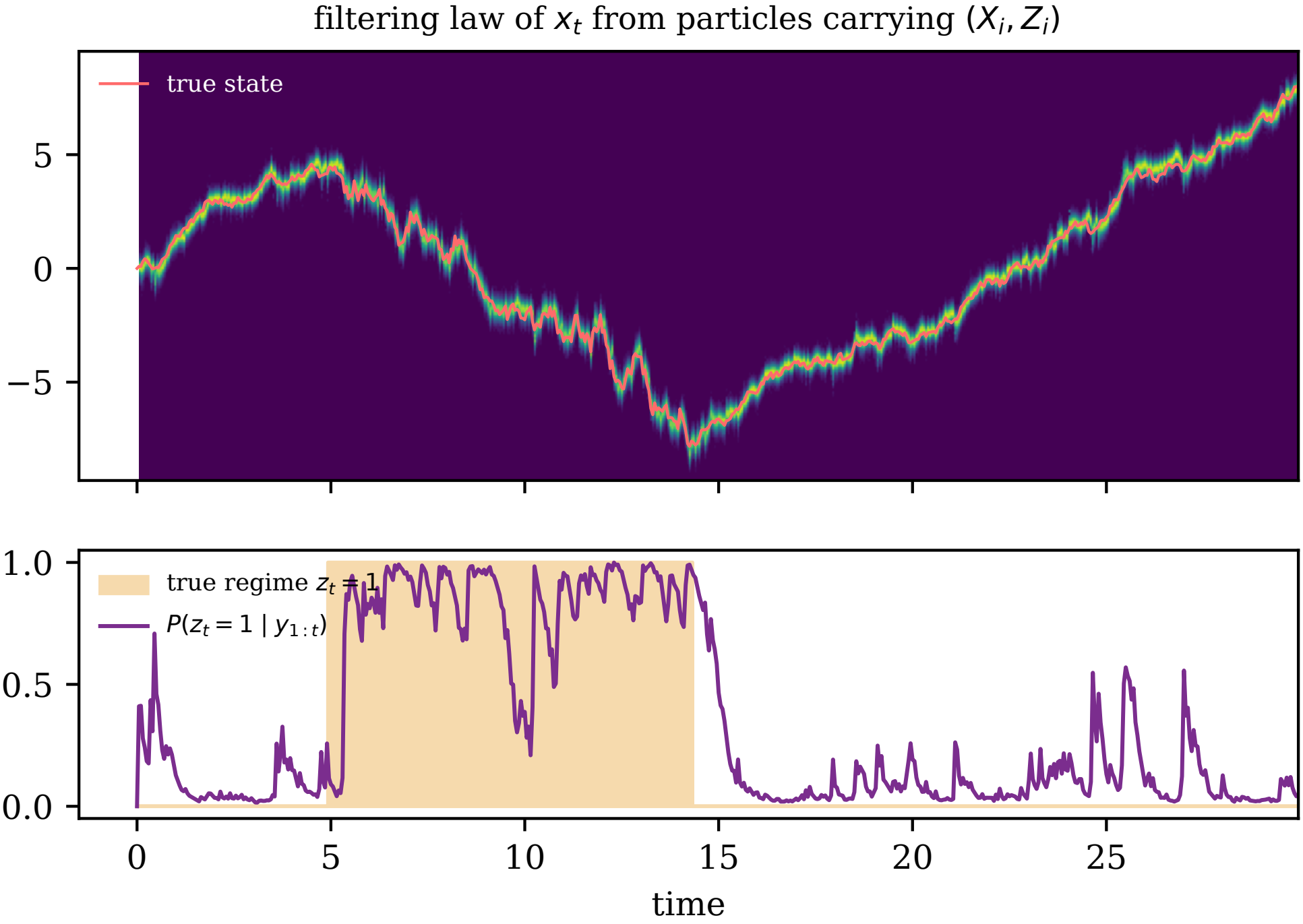

filtering law of x_t from particles carrying (X_i, Z_i)
true state
5
0
−5
true regime $z_t = 1$
$P(z_t = 1 \mid y_{1:t})$
1.0
0.5
0.0
0
5
10
15
20
25
time